

Yoga for the Grid - Flexible interconnection for utility scale and distributed generation

By

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Introduction

The US needs new clean electricity generation to lower greenhouse gas emissions and meet rising demand from electrification of transportation, buildings, and industry as well as data centers. Despite the needs, it is shockingly difficult and expensive for new generation to actually connect to both the transmission and distribution grids in most of the United States. Texas's grid operator Electric Reliability Council of Texas (ERCOT) leads the country in deploying renewables and batteries with its connect & manage approach to interconnection, averaging while the rest of the country uses an invest & connect approach that slows and stops projects. We consider non-guaranteed grid capacity and a transmission or distribution operator's ability to adjust and curtail the output of a generator in anticipation of or response to system needs as flexible interconnection. Though technologies exist to manage flexible interconnection of generation the policy is not in place to enable it.

Problem Statement

In most of the U.S. grid outside of ERCOT, the generator interconnection process creates a major bottleneck for deploying new renewable energy and storage resources. Transmission operators generally use the Network Resource Interconnection Study (NRIS) framework to study whether projects can deliver power to load under worst-case system conditions, and assign the full cost of any network upgrades to the project. This project-by-project, upgrade-driven approach frequently triggers withdrawals when developers cannot absorb those costs, which in turn forces restudies for remaining projects and extends delays across the queue. A parallel problem exists for distribution grid interconnection, where slow processes, lack of flexibility, and lack of data prevent easy interconnection. In a world of "smart grids" that take advantage of flexibility and

distributed resources, this is a non-smart approach to interconnection. This research examines the possibilities and limitations of bringing a flexible approach to interconnecting generation.

Methodology

The team conducted a review of the academic and grey literature on flexible interconnection of generation at the transmission and distribution levels and the tools needed to implement flexible interconnection. Literature was found through searches in academic journal databases, web searches, and AI queries for resources. Recommendations from different sources were compiled and evaluated. The team also conducted interviews with subject matter experts as detailed in Appendix A, which were necessary to ensure the research captures the complexities of managing transmission and distribution systems.

Results

Flexible Interconnection of Utility Scale Generation

Transmission interconnection requests escalated by three to five times across the U.S. between 2014 and 2024, and the growth of renewables will likely continue that trend (Gorman et al., 2024). There are two main interconnection strategies for utility scale generation in the US. ERCOT uses connect & manage, where generators only pay for the infrastructure needed to connect to the grid, and accept risk of redispatch and curtailment under congested transmission. A related feature of ERCOT's model is its reliance on proactive transmission planning and congestion management, rather than relying solely on generator-driven upgrade triggers (Howland, 2023; Driscoll, 2023). The rest of the US uses invest & connect, which guarantees

deliverability of power even during congestion, allowing generators to count towards resource adequacy. This is called a Network Resource Interconnection Service (NRIS), which comes with high time and monetary costs when developers must pay the full transmission system upgrades required. Systems that use NRIS also have Energy Resource Interconnection Service (ERIS) options which function more like connect & manage, such that the generator only has access to transmission “as available” but doesn’t have to pay or wait for major grid upgrades (Norris, 2023, p. 3). However, ERIS is often still embedded in NRIS study processes and timelines that can make it much slower and less predictable than ERCOT’s connect-and-manage model (Federal Energy Regulatory Commission, 2023, pp. 2–5; Norris, 2024, pp. 2–6).

Though researchers and reporters express hope in ERIS as a solution to interconnection challenges (Norris, 2023; Engermeier, 2023; Howland, 2023), Grid Strategies’ report based on interviews of interconnection customers as well as subject matter experts Joshua Keeling and Jennifer Galaway pointed to the fact that most projects choose NRIS and as ERIS projects face distinct barriers (Wilson et al., 2024). In order to make ERIS an effective solution to interconnection queues those barriers must be addressed.

Technologies and Tools to Manage Congestion

There are a number of different technologies and tools that help grid operators manage congestion and different market structures may place more emphasis on some tools than others. ERCOT’s energy only market leans more heavily on operational tools such as redispatching, curtailment and congestion pricing.

Redispatching refers to the process by which a transmission system operator (TSO) changes the power allocations of power plants...to maintain the balance between electricity generation and consumption in the grid and to avoid bottlenecks. (Kyon Energy)

This process is facilitated by Security-Constrained Economic Dispatch (SCED), a widely used software across both ERCOT and the ISOs/RTOS. The SCED software evaluates real-time market data such as available generation, load levels and generator offers to determine the least cost dispatch of resources while balancing other constraints such as transmission limits, frequency/voltage reliability needs, generator ramp up rates and outages.

All seven major ISOs/RTOs use redispatching in day to day operations to manage congestion. It can be thought of as a “baseline” tool where dispatchable resources, such as natural gas plants, hydropower and battery storage can be “moved up or down” to relieve transmission constraints. For example, if a line is overloaded, the system operator can reduce output from one generator and increase output from another generator located on the other side of the constraint.

In ERCOT, SCED runs every five minutes to produce least-cost dispatch instructions as well as calculating prices for different locations throughout the transmission system. These prices are known as Locational Marginal Prices (LMP). (ERCOT) LMP sends accurate price signals to generators and customers, informing them when and where power is cheap or expensive. (Ozkan, 2022)

LMPs vary by location and consist of three main pieces: energy, congestion and losses. The congestion component reflects the added cost of serving load at a given location because the system had to redispatch power around a constraint/bottleneck in the transmission system in order to serve demand. (ISO New England)

These added costs serve as signals that reveal where there is congestion in the grid and thus can help identify where upgrades to transmission are needed or where GETs, storage or other operational tools would be most beneficial.

Another cornerstone tool in ERCOT's toolbox is curtailment– the intentional reduction of power generation, usually from wind, solar, or other renewable sources when there is more electricity being produced than the grid can safely handle. (Dutton, 2024)

There are two kinds of curtailment: economic and reliability. Under economic curtailment, a generator could reduce output because of market signals such as energy prices becoming too low, making it uneconomically viable to keep operations going.

On the other hand, reliability curtailment is responsive to physical constraints such as transmission congestion, thermal line limits and outages. Generators must be curtailed in order to maintain safety and reliability across the power system.

An example of reliability curtailment could be a scenario with a solar facility capable of producing 200 MW, but the transmission line near it is congested and can only safely carry 150 MW. Grid operators then instruct the solar facility to reduce, “curtail” its output by the amount of MW necessary to stay within the safe operating limits.

Under ERCOT's connect-and-manage program, new generators take on this curtailment risk in exchange for being able to interconnect and begin operations sooner. During periods when the grid is unconstrained, the generator can produce power at full capacity and benefit from the financial gains. But during times of congestion, it must accept output limits determined by ERCOT.

Grid-Enhancing Technologies

Three of the best known Grid-Enhancing Technologies (GETs) are dynamic line ratings, advanced power flow control and topology optimization. These hardware and software tools help grid operators make better use of the existing transmission system by identifying additional line capacity, redirecting power flows, or reconfiguring the grid to relieve congestion. (Watt Coalition)

In ERCOT, GETs support the connect-and-manage approach by alleviating transmission constraints through operations after generators have interconnected. In ISOs/RTOs, GETs can also help speed up interconnection, but must work within the confines of market structures with structure requirements for deliverability.

Dynamic Line Ratings (DLR), another type of GET, offer insights into the actual capacity of a transmission line throughout the day as the temperature fluctuates and wind patterns change. This matters given power lines' sensitivity to temperature. For example, as the wind blows this can cool down conductors and increase the amount of power a line can carry.

DLR is particularly well suited to manage congestion on older lines and while new lines may be designed to avoid a thermal limit, use of DLR can still be beneficial by providing situational awareness and supporting asset management, according to a report by the Department of Energy. (U.S. Department of Energy, 2019)

This additional situational awareness gives operators more insight into the grid's actual real-time capacity which matters as they determine whether a renewable generator needs to be curtailed or not and by how much.

This technology has been tested in ERCOT where Oncor installed monitoring equipment on several transmission lines experiencing congestion in the western part of the state. Using SCED,

ERCOT was able to incorporate the line ratings into their software and improve their resource dispatching. (Oncor, 2013)

One important regional consideration is that DLR is most useful when the constraint on a transmission line is thermal capacity. Constraints can differ across the country as each region has its own set of weather patterns that put different pressures on the grid. If challenges in a territory center more on voltage stability, system stability or transformer limits, as opposed to thermal limits, then increasing the line's dynamic rating may not significantly increase capacity.

Topology optimization is another GET that is like the grid's navigation app – when there's too much traffic on one power line, it can find alternative, reliable routes for the power. (Watt Coalition)

One example of such a tool is NewGrid, a software that quickly finds and evaluates reliable reconfigurations to reroute flow around congestion. (Ruiz, 2025)

By switching circuit breakers on and off, this software is able to reconfigure how power flows through power lines, leveraging operations to help relieve congestion and maximize existing infrastructure without building new transmission.

In a 2025 presentation to ISO-New England's Planning Advisory Committee forum on GETs, NewGrid reported that its topology optimization software reduced the most heavily loaded constraint flows caused by planned outages by 31% on average across cases where promising solutions were identified.

For ISOs and RTOs considering their own version of ERCOT's interconnect-and-manage approach, this type of tool could help manage post-interconnection congestion more dynamically. This kind of tool alone cannot alone solve all congestion issues, but it can

complement other tools from the congestion management suite such as redispatching, curtailment and dynamic line ratings.

Uptake of these tools: Energy-Only Markets vs Capacity Markets

Energy-only markets like ERCOT optimize for fast interconnection. Generators are able to interconnect quickly, but take on the risk of curtailment whereas capacity markets optimize for deliverability and reliability. Since ERCOT is located solely within the state of Texas and not subject to meet certain FERC requirements like capacity markets are.

Capacity markets have more of an invest-and-connect approach. A capacity market does not pay for the energy produced but instead pays for the ability to produce power when needed. (Federal Energy Regulatory Commission)

In other words, they are designed to pay resources for being available to meet future reliability needs and help meet peak demand under stressed conditions, not just for producing energy in real time.

Utility incentive - misalignment

An important nuance when considering uptake of these various technologies is structural challenges that these various low-cost tools can face within capacity markets, even when they prove to reduce congestion in meaningful ways.

While low-cost software tools may reduce congestion, they likely won't produce the kinds of earnings as large capital investments like transmission projects. As a paper on dynamic line ratings by the Department of Energy states, "the U.S. provides transmission owners little

incentive to deliver more power over existing lines or to reduce transmission congestion.” (U.S. Department of Energy)

Congestion in ISO-NE region

An additional consideration is how much congestion a region actually has as this could be even less incentive to adopt some of the technologies described so far such as in ISO-NE where grid congestion is low.

According to the 2024 Assessment of the ISO New England Electricity Markets by Potomac Economics, the ISO-NE region experiences less congestion than other RTOs, “averaging \$0.33 per MWh over the past three years. In contrast, congestion levels per MWh of load in the other RTOs were six to thirteen times higher. The low level of congestion in New England can be attributed to substantial investments in transmission infrastructure made over the past 20 years to address reliability needs and support local transmission security.” (Potomac Economics, 2025)

Discussion

Recommendations to Accelerate Flexible Interconnection of Generation outside ERCOT

Outside ERCOT, the most likely path to accelerating interconnections through a connect & manage style approach is making ERIS an appealing option to interconnection customers. This can be done by improving speed, reducing costs, and ensuring predictability (Norris, 2024, pp. 2–6; U.S. Department of Energy, 2024, pp. 18–24). The recommendations below therefore focus

on changes to transmission planning and operations and interconnection studies that could make ERIIS a more effective tool for reducing interconnection queue delays.

Transmission Planning and Operations

1. **Interconnect and dispatch renewable resources paired with storage** that can take advantage of the spongy time-machine nature of batteries to become dispatchable resources and continue generating even when their grid output is curtailed. The main challenge with ERIIS is the risk of curtailment. Battery storage can help solve the curtailment challenge in two ways. Onsite batteries at the point of interconnection can soak up extra power that would otherwise be curtailed and shift it to times when the grid is unconstrained. By adding flexibility to load batteries can also shift demand to align with variable generation, for example when solar is at full production (Idaho National Lab, 2024)
2. **Use proactive transmission planning** to account for forecasted load growth and the ability to meet it with deliverable clean generation and storage (U.S. Department of Energy, 2024, pp. 26–32). Interviews with subject matter experts made clear that proactive transmission planning is a key enabling condition for successful implementation of connect & manage or accelerated ERIIS interconnection. Proactive transmission planning removes risk from developers, accelerates interconnection opportunity, and allows for the development of significant renewable resources. ERCOT demonstrated the success of this approach with the success of its Competitive Renewable Energy Zones initiative that delivered over 18 GW of new wind generation (Americans for a Clean Energy Grid, 2017). However, CAISO is the only RTO successfully using this model today (Wilson et al., 2024).

3. **Transmission operators make publicly available useful data** on transmission utilization and capacity, potential constraints, and likely upgrade needs so developers can make better ERIS siting and interconnection decisions (Gorman et al., 2024, pp. 20–21).

In interviews both Clark Stephens and Josh Keeling emphasized the importance of data to interconnection customers in their ability to model ERIS risks and upsides. In much of the west the system is especially opaque given the lack of market pricing and dispatch. Only once interconnection customers have data to make the case for the benefits of ERIS will that be a preferable interconnection approach over NRIS.

4. **Transmission operators and utilities join an energy market to efficiently dispatch resources across the transmission system. Ideally pick one market for the west.**

Well-functioning energy markets are purpose built to select the least-cost portfolio of resources while accounting for transmission system constraints. With the appropriate financial mechanisms in place, ERIS resources bidding into a market should be selectable when transmission capacity exists, and even during congestion when an NRIS resource would be more expensive to dispatch.

Interconnection Studies & Requirements

5. **Separate and sequence ERIS studies from NRIS studies and transmission upgrades.**

Several proposals preceding and following FERC Order No. 2023 have argued that ERIS and NRIS should be studied separately so that projects seeking less-than-fully-deliverable service are not forced through the same study pathway as projects seeking full deliverability (Norris, 2024, pp. 4–6). The goal is to shorten the queue and lower costs for

ERIS projects to interconnect while still allowing them to proceed with NRIS to improve their finances and capacity contribution to the grid.

6. **Identify clear timelines for ERIS studies.** Transmission operators should establish clear study timelines and queue management practices so developers face less uncertainty and financing risks (Gorman et al., 2024, pp. 18–21). To the extent that ERIS studies can be automated with AI and/or machine learning those tools should be used to accelerate the process.
7. **Utilize non-wires alternatives to enable interconnection.** The U.S. Department of Energy has identified case studies where the grid operator identified an expensive rebuild or new transmission need, while the interconnection customer found a cheaper non-wires solution (Gorman et al., 2024, p. 17). Finding cheap and fast solutions improves interconnection queues and energy costs. Grid Strategies identified CAISO as leading the nation on using alternatives to full interconnection (Wilson et al., 2024, p. 61). Grid congestion management tools that support fast interconnection already exist, but given fundamental differences in market structures, approaches like ERCOT’s connect-and-manage may not work well within capacity markets where there is a stronger focus on deliverability and reliability and where strong incentives to build large capital projects may not favor adoption of lower cost options. However, these tools can still be leveraged by capacity markets to improve and facilitate faster interconnection for new resource generators.
8. **Transmission operators model likely timing and frequency of transmission constraints during interconnection studies and share the information.** Modeling the timing of constraints and beyond just worst-case conditions allows developers to better

assess expected curtailment risk (Norris, 2023, p. 63). Providing project-specific information about when and why curtailment may occur can improve developer decision-making and better target study resources toward the issues that matter most for a flexible interconnection decision. This should happen proactively in transmission planning and data sharing, as well as part of interconnection studies.

Policy Needs for distribution utilities to allow flexible interconnection of DERs

Legislative and Regulatory Mandates

As NVIDIA's product roadmap anticipates single racks consuming 1MW and new data center projects in South and Southwest demand of at least 2GW, the traditional "invest and build" utility model has become obsolete. The One Big Beautiful Bill Act passed in 2025 in which it curtailed clean energy tax credits and shifted the political and economic focus toward extreme cost-efficiency. Due to this push and high-interest, utilities must transition to a "connect and manage" strategy. This will then transition DERs from a passive edge asset to a core reliability resource, ensuring existing infrastructure is maximized before ratepayers are burdened with trillions in capital expenditures for centralized plants and transmission.

State-level mandates are increasingly codifying this shift by creating market certainty and competitive accountability. For example, Virginia's H.B. 2346 requires utilities to implement Virtual Power Plant capacity of up to 450MW. This mandate will allow for third-party aggregators to compete directly with utility resources to drive innovation and lower costs.

In order to execute this shift, the follow Policies are required

- 1) VPP Capacity Targets: Establishing specific megawatt goals creates the market demand necessary for developers to invest in local infrastructure (Pew Charitable Trusts, 2026).
- 2) Regulatory Oversight: Independent monitors are essential to oversee solicitations and verify performance, ensuring third-party providers are not marginalized by utility resources.
- 3) Reliability standards: Adoption of NERC Level 2 industry recommendations for large load interconnections is non-negotiable. Having these standards prioritize proactive operational readiness and disturbance recovery for massive new loads (Clarion Energy, 2025).

Integrated Planning and Data Transparency

To scale DERs, utilities must adopt integrated system planning (ISP), which treats DERs as a “foundational building block” in long-term resource forecasts. The strategic utility must utilize Hosting Capacity Analysis (HCA) to inform developer siting.

To establish this foundation, the following should be implemented:

- 1) HCA Accessibility: Having utilities required to publish unredacted hosting capacity data without restrictive NDA’s. Developers should have the transparency needed to avoid constrained areas and propose viable projects (Driscoll, 2025).
- 2) Integrated Resource Planning (IRP) Reform: Regulators must mandate that IRPs utilize bottom-up modeling to predict customer led DER adoption rather than relying on outdated top-down uptake rates (Pew Charitable Trusts, 2026).
- 3) Hosting Capacity Tools: Frequently updated tools must model the impact of multiple DER technologies simultaneously to reflect real grid interactions (Baldwin et al., 2025).

Financial Alignment and Incentives

Today we use a “Cost-Causer-pays” model which is not suitable to modern day and kills viable projects. Furthermore, VPPs and DERs deliver power during peak demand at 40% to 60% of the cost of traditional infrastructure (Pew, 2026), yet utility motives remain biased toward capital focused hardware(CAPEX). To bridge the gap, the following should be enforced:

- 1) Performance-Incentive Mechanisms (PIMs): Utilities must be rewarded or penalized based on interconnection speed and efficiency.
- 2) Shared Savings Mechanisms: Utilities must be rewarded or permitted to retain a portion of the savings generated by choosing Non-Wires Alternatives(NWA) over traditional upgrades. The standard for this is a 30% utility/70% customer split, providing a clear pathway to profit from software solutions (Pew, 2026).
- 3) Cost allocation Reform: Cost should be distributed among all who benefit from the upgraded circuit. This can include building a reserve fund from interconnection fees to finance subsequent upgrades (Baldwin et al., 2025).

Technical Design and Standard Adoption

Utilities must move from “firm” connections to “conditional” or “flexible” agreements. The technical bedrock is IEEE 1547, which must be utilized not just for safety, but as a tool for voltage control and disturbance recovery. Additionally, while data centers can shift compute loads, the use of onsite generation is a more broadly applicable technical strategy for managing peak stress across all ownership types(Clarion, 2025).

Having a Distributed Energy Resource Management System (DERMS) offers the highest grid utilization, though simpler autonomous systems remain viable for predictable loads.

Streamlined and Automated Process

“Soft costs” can reach 78% of total project costs for residential DERs, largely due to administrative friction(Pew, 2026). Automation is the only way to manage the volume of requests required to meet electrification targets.

- 1) Permitting Automation: Platforms like SolarAPP+ reduce staff review time and standardize safety checks. Texas has successfully utilized third party licensed professionals and automated platforms to issue permits within three business days.
- 2) Queue Management: The Connect-and-Manage (C&M) model, as seen in ERCOT, outperforms the traditional NRIS approach. By bypassing exhaustive studies and managing congestion via dispatch, ERCOT achieved a 22% completion rate, compared to just 14-18% in other ISOs (Angermeier, 2026).
- 3) Automated Application Review: The industry standard for scale is the UK Power Networks “Smart Connect” portal, which handles 60,000 annual requests and approves 80% within minutes (Pew, 2026).

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Appendix A: Subject Matter Expert Interviews

Transmission Interconnection SMEs:

- Clark Stephens, The Nature Conservancy, Senior Policy Advisor
- Joshua Keeling, Utility API, Chief Commercial Officer
- Jenniver Galaway, Portland General Electric, Senior Manager, Transmission Strategy & Regulation

Transmission Interconnection Guiding Questions for 30-45 minute interviews

1. How might transmission operators outside ERCOT implement flexible transmission interconnection ala connect and manage or expedited ERIS interconnection? What about transmission operators outside an RTO/ISO?
2. How would the participation in a day-ahead market change the ability of a transmission operator to implement flexible interconnection?
3. How would flexible interconnection change project finance for renewable energy projects?
4. How does being FERC jurisdictional or non-FERC jurisdictional affect a transmission operator's ability to unilaterally decide to implement flexible interconnection?